NMC 2015 PRIMARY LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. $9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 = 9^9 \times$ ___

Qn2. If $\frac{11}{7} - \frac{2}{3} = \frac{x}{19}$ then $x =$_____

Qn3. A square whose side is 16cm has the same area as a rectangle whose side is 8cm. Find the perimeter of the rectangle.

Qn4. Given that $a * b = a^2 + b^2$, then $(-2)^2 * 2^2 =$ ___

Qn5. A certain positive number is divisible by 3 and also by 5. When the number is divided by 7, the remainder is 4. What is the smallest positive number that satisfies these conditions?

Qn6. A Jar filled with water weighs 10 kg. When half of the water is poured out, the jar and remaining water weigh 5.75 kg. What is the weight of the jar?

Qn7. If a is divided by b, the result is $\frac{2}{3}$. If b is divided by c, the result is $\frac{7}{8}$. What is the result when a is divided by c?

Qn8. A debating club is made up of pupils from P5, P6 and P7 only. Five pupils are from P6, Ten pupils are from P7, and one-sixth of the group are from P5. How many pupils make up the club?

Qn9. Joana stands in a queue of pupils at the morning assembly. She is the 26th pupil counting from the front of the queue. She is the 15th pupil counting from the end. How many pupils are in the queue?

Qn10. Thirteen oranges weigh as much as two mangoes and one pawpaw. Four oranges and one mango have the same weight as one pawpaw. How many apples have the same weight as one pawpaw?

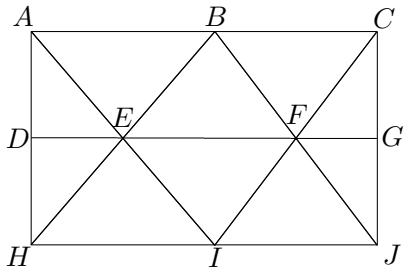
SECTION B

Qn11. Mrs. Kafeero went on a round trip from town X to town Y. The distance between town X and town Y is 640 km. The cost of petrol is Shs. 3500 per litre. The performance of her car is 12.8 km per litre.

(a) How much will the trip cost her?

(b) Twenty minutes into the trip, she got a flat-tyre and it took her 30 minutes to fix it. How much more fuel did she have to add?

Qn12. Given



(a) Write down all the triangles in the figure (for example: BHJ).

(b) How many are they?

Qn13. Rose sent her son to a supermarket with a Shs. $20,000$ note. The son bought the following items: s kgs of sugar at Shs. 2600 each; b fingers of bananas at two for Shs. m. Write down an expression for the amount of money that remained unspent after buying all the above.

Qn14. Arrange the Roman Numerals I, II, III, IV, and V in order from left to right using the following information:

(i) I must be before III but after IV (ii) II must be after IV but before I
(iii) V must be after II but before III (iv) V must not be the third numeral

Qn15. A bus can carry 55 seated passengers. The bus starts out empty and picks up 1 passenger at the first stop, 2 passengers at the second stop, 3 passengers at the third stop, 4 passengers at the forth stop, and so on (picking n passengers at the nth stop).

(a) After how many stops will the bus be full?

(b) Once the bus is full, the driver decides to allow standing passengers. If he picks at only one more stop but still following the same pattern of picking n passengers at the nth stop, how many passengers will be standing in the bus?

(c) If every passenger pays Shs. 1000 irrespective of the point at which they boarded the bus and whether they are seated or standing, how much money will the bus conductor collect?

NMC 2015 PRIMARY LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1. $9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 + 9^9 = 9^9 \times 9$.

Qn2. $\frac{19}{21} = \frac{x}{19}$. By cross multiplication we get $19 \times 19 = 21 \times x$. Therefore $x = \frac{19 \times 19}{21} = 17.19$.

Qn3. Area of square of side length 16cm is equal to $16 \times 16 = 256\text{cm}^2$.
Let l be the length of the rectangle. Then its area A, is equal to $A = l \times w = l \times 8 = 8l\text{cm}^2$. Since the area of the square is equal to the area of the rectangle, it implies $8l = 256 \implies l = \frac{256}{8} = 32\text{cm}$. The perimeter of the rectangle $p = 2(l + w) = 2(32 + 8) = 2 \times 40 = 80\text{cm}$.

Qn4. $(-2)^2 * 2^2 = 4 * 4 = 4^2 + 4^2 = 16 + 16 = 32$.

Qn5. Multiples of 3 and 5 are $15, 30, 45, 60, 75, 90, \cdots$. When each of these numbers is divided by 7, the smallest number to leave a remainder of 4 is that smallest number that is 4 more than a multiple of 7. Of these numbers, the required number is 60.

Qn6. Let the weight of the jar be x kg and the weight of water be y kg. Then

$$x + y = 10 \tag{1}$$

$$x + \frac{y}{2} = 5.75 \tag{2}$$

Eqn(1)- Eqn(2) gives $\frac{y}{2} = 4.25$. Therefore $y = 2 \times 4.25 = 8.5$ kg. Hence $x = 10 - 8.5 = 1.5$. Therefore the weight of the jar is 1.5kg.

Qn7. $\frac{a}{b} = \frac{2}{3} \Longrightarrow a = \frac{2}{3}b$. Also $\frac{b}{c} = \frac{7}{8} \Longrightarrow c = \frac{8}{7}b$. Therefore $\frac{a}{c} = \frac{\frac{2}{3}b}{\frac{8}{7}b} = \frac{2}{3} \div \frac{8}{7} = \frac{2}{3} \times \frac{7}{8} = \frac{7}{12}$.

Altrenatively $\frac{a}{c} = \frac{a}{b} \times \frac{b}{c} = \frac{2}{3} \times \frac{7}{8} = \frac{14}{24} = \frac{7}{12}$.

Qn8. Let x be the size of the team. Then the size of P5 class is $\frac{x}{6}$ and therefore $\frac{x}{6} + 5 + 10 = x \Longrightarrow \frac{x}{6} + 15 = x$. Thus $\frac{x}{6} - x = -15 \Longrightarrow \frac{-5}{6}x = -15$. So, $x = \frac{15 \times 6}{5} = \frac{90}{5} = 18$.

There are 15 pupils in P5 class.

Qn9. There are 25 pupils ahead of Joana and 14 pupils behind Joana. Therefore there are 25+1+14 = 40 pupils in the queue.

Qn10. Let a represent apples, m, the mangoes and p the pawpaw. Then

$$13a = 2m + p \tag{3}$$

$$p = 4a + m \tag{4}$$

Substituting for p in the Eqn(3), $13a = 2m + 4a + m \Longrightarrow 13a = 3m + 4a$ so that $3a = m$. Substituting for $m = 3a$ in the Eqn(4) gives $4a + 3a = p \Longrightarrow p = 7a$. Therefore 7 applies have the same weight as one pawpaw.

SECTION B

Qn11. (a) Total distance, $D = 640 \times 2 = 1280$km.

The total number of litres used, $L = \frac{1280}{12.8} = 100$litres.

The total cost of her trip= cost per litre × number of litres used $= 3500 \times 100 = 350,000$.

The total cost of Mrs. Kafeero's trip is $Shs.350,000$.

(b) There is no relationship between the amount of fuel used and the time taken when in the case of a flat-tyre or they are 0 litres of fuel used.

Qn12. (a) The triangles are $ABE, ACI, AED, AEH, ABH, AIH, BCF, BCJ, BFE, BJH, CGF, CJF$ $CJI, DEH, EIH, EFI, FGJ, FJI$.

(b) There are 18 triangles.

Qn13. Sugar cost Shs. $2600s$ while bananas cost Shs. $\frac{bm}{2}$.

Therefore the total mount of money spent is Shs. $2600s + \frac{bm}{2}$. The amount of money unspent

after buying the items is Shs.$\left(20000 - (2600s + \frac{bm}{2})\right) = Shs.\left(20000 - 2600s - \frac{bm}{2}\right)$.

Qn14. The arrangement from left to right is IV, II, I, V, III.

Qn15. (a)

Stop number	1	2	3	4	5	6	7	8	9	10
Passengers picked	1	2	3	4	5	6	7	8	9	10
Passengers in the bus	1	3	6	10	15	21	28	36	45	55

(b) 11 passengers will be picked at the 11th stop.